

Introduction to Macroeconomics

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Problem Set 2

2026

1 Consumer Price Index

Exercise 1 (problem 2, chapter 24, Mankiw-Taylor)

Suppose that the residents of Vegovia spend all of their income on cauliflower, broccoli and carrots.

- In 2004 they buy 100 heads of cauliflower for EUR 200, 50 bunches of broccoli for EUR 75 and 500 carrots for EUR 50.
 - In 2005 they buy 75 heads of cauliflower for EUR 225, 80 bunches of broccoli for EUR 120 and 500 carrots for EUR 100.
1. If the base year is 2004, what is the CPI in both years ?
 2. What is the inflation rate in 2005 ?

Exercise 2 (Exam June 2013)

Assume that in 2010 an employee earns €2500. In 2011 her nominal salary increases by €200. Taking 2010 as the base year, the CPI in 2011 is 108. By how much did the real salary of the employee increase ?

- (a) 2%.
- (b) 3%.
- (c) 0%.
- (d) 1%.

Exercise 3

Suppose we have developed a price index, the College Student Price Index (CSPI), which is based on a survey of annual purchases of a representative college student. Table below gives information regarding the consumption basket on which the CSPI is based and the prices of each good in 2000 (base year), 2001 and 2002.

Good	Quantity	2000 (Euros)	2001 (Euros)	2002 (Euros)
Notebooks	10	2	1	3
Calculators	1	50	60	70
Large Coffees	200	1	1	1
High-Caffein Sodas	50	2	3	4
Textbooks	10	100	120	150

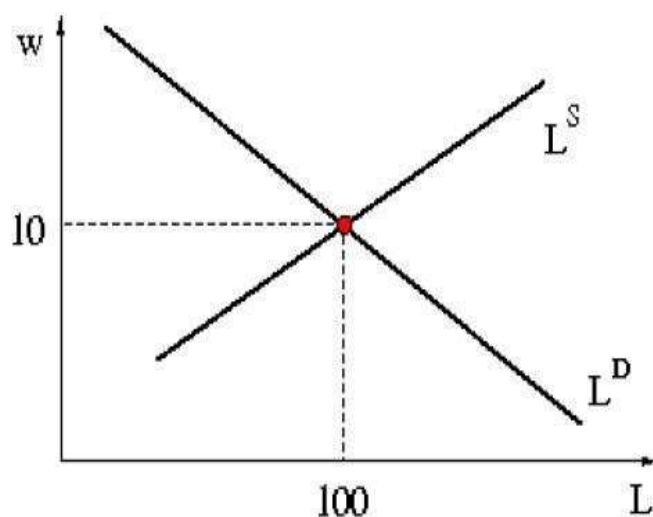
In 2000, calculators selling for 50 euros were able to perform a variety of mathematical functions. In 2001, the price of calculators rose to 60 euros, but had color displays and could easily be programmed. In 2002, calculators sold for 70 euros and came with additional memory sticks and the ability to link with personal computers. Which of the following statements is true?

- (a) The CSPI underestimates the cost of living for students.
- (b) The CSPI accurately estimates the cost of living for students.
- (c) The CSPI overestimates the cost of living for students.
- (d) We do not have enough information to reply to this question.

2 Labor market

Exercise 4

Consider the graph below, which represents the equilibrium in the labor market :



Which of the following statements is correct ?

1. The marginal productivity at the equilibrium is equal to 10.
2. If a minimum wage is set below 10, unemployment will not appear.
3. Only the 100th worker earns the value of his/her own reservation
4. All of the above statements are correct.

Exercise 5

The following table presents employment data for a hypothetical nation (Chomaland) for 2009 and 2010.

	2009	2010
Unemployment Rate	6.4%	6.2%
Number of Employed Workers	137,738,000	137,478,000

Suppose the government of this country reported that the unemployment rate has fallen. However, the number of employed workers decreased by 260,000 between 2009 and 2010. Which of the following is a possible explanation for this phenomenon ?

1. The number of unemployed persons decreased from 2009 to 2010. If enough unemployed workers drop out of the active labour force, the unemployment rate can fall even if the number of employed workers decreases.
2. Size of the labour force increased from 2009 to 2010. If the size of labour force increases while the number of employed workers falls, the unemployment rate also falls.
3. The government has made a mistake. It is not possible for both the unemployment rate and the number of people employed to decrease simultaneously.
4. None of the explanations above is a good explanation for the phenomenon that has been observed.

Exercise 6

Let's assume that in a given country the productivity falls down suddenly, which has a negative impact on the production function.

1. What happens to the labor demand curve ?
2. What is the impact of this change in productivity on the labor market (on the employment level and the real wage), assuming that the labor market is always in equilibrium ?
3. What is the impact of this change in productivity on the labor market if the unions are not allowing the real wages to be decreased ?

3 Additional exercises (will not be discussed in the seminar)

Exercise 7 (Exam June 2007)

Suppose that your income increases from 19'000 CHF to 31'000 CHF while simultaneously the Consumer Price Index (CPI) increases from 122 to 169. Your purchasing power :

- (a) decreases
- (b) increases
- (c) remains constant
- (d) you cannot reply to this question without knowing the base year

Exercise 8 (Exam in June 2013)

Which of the following types of unemployment is caused by the fact that unemployed workers need time to get information about available job vacancies on the market ?

- (a) Seasonal unemployment
- (b) Structural unemployment
- (c) Cyclical unemployment
- (d) Frictional unemployment

Exercise 9 (Exam August 2014)

Structural unemployment **does not depend on** :

- (a) the setting of efficiency wages by companies.
- (b) the introduction of a minimum wage at the national level.
- (c) seasonal changes in labor demand from businesses.
- (d) the strength of labor unions in collective bargaining agreements.